

B3: Atomic and Laser Physics — MT 2020 Model Solutions

Question 1 — Central field, Thomas–Fermi atom

Problem. For an N -electron atom: (a) state the central-field approximation and the good quantum numbers; (b) sketch the effective potential of sodium; (c) using a Fermi-gas description ($n = 8\pi p^3/3h^3$) derive the Thomas–Fermi equation $\chi'' = \chi^{3/2}/x^{1/2}$ with $x = r/b$ and find b ; (d) discuss its limits; (e) explain why $4s$ fills before $3d$ in K.

(a)

The two-body term $\sum_{i>j} e^2/(4\pi\epsilon_0 r_{ij})$ prevents separation. Replace by spherical mean acting on each electron:

$$\sum_{i>j} \frac{e^2}{4\pi\epsilon_0 r_{ij}} \longrightarrow \sum_i S(r_i).$$

Each electron sees a central potential:

$$U(r) = -\frac{Ze^2}{4\pi\epsilon_0 r} + S(r).$$

Hamiltonian becomes a sum of one-electron terms; orbitals are labelled by single-particle quantum numbers:

$$\boxed{n, \ell, m_\ell, m_s}$$

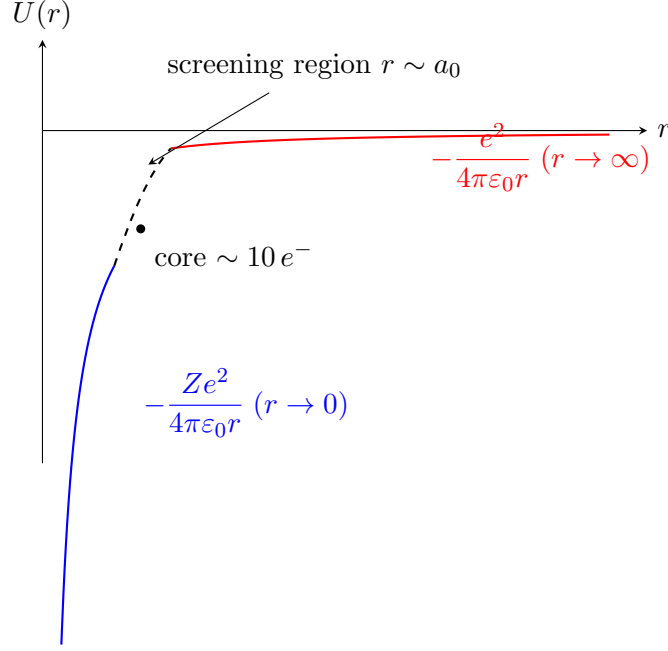
Energies depend on n and ℓ (non-Coulombic). Residual electron repulsion couples individual ℓ_i, s_i so only L, S, J, M_J are strictly good for the full atom.

(b)

Two limits set the shape of $U(r)$ for Na ($Z = 11$, 10 core electrons):

$$\begin{aligned} r \rightarrow 0 : \quad U(r) &\rightarrow -\frac{Ze^2}{4\pi\epsilon_0 r}. \\ r \rightarrow \infty : \quad U(r) &\rightarrow -\frac{e^2}{4\pi\epsilon_0 r}. \end{aligned}$$

Screening turns on over $r \sim a_0$.



(c)

Local Fermi momentum from $p_F^2/2m = -eV(r)$ (electron just bound, $V(\infty) = 0$):

$$n_e(r) = \frac{8\pi}{3h^3} [-2meV(r)]^{3/2}.$$

Write $V(r) = \chi(r)Ze/(4\pi\epsilon_0 r)$, $Z_{\text{eff}} = \chi Z$, BCs $\chi(0) = 1$, $\chi(\infty) = 0$. Poisson's equation away from nucleus:

$$\nabla^2 V = \frac{1}{r} \frac{d^2(rV)}{dr^2} = \frac{e n_e(r)}{\epsilon_0}.$$

Substitute $rV = Ze\chi/(4\pi\epsilon_0)$:

$$\frac{Ze}{4\pi\epsilon_0} \frac{1}{r} \frac{d^2\chi}{dr^2} = \frac{e}{\epsilon_0} \frac{8\pi}{3h^3} \left[\frac{2me^2 Z\chi}{4\pi\epsilon_0 r} \right]^{3/2}.$$

Isolate χ'' :

$$\frac{d^2\chi}{dr^2} = C \frac{\chi^{3/2}}{r^{1/2}}, \quad C = \frac{32\pi^2}{3h^3} \frac{(2m)^{3/2} e^2 Z^{1/2}}{(4\pi\epsilon_0)^{3/2}}.$$

Set $r = bx$:

$$\frac{1}{b^2} \frac{d^2\chi}{dx^2} = C \frac{\chi^{3/2}}{b^{1/2} x^{1/2}}.$$

Demand the universal form $\chi''(x) = \chi^{3/2}/x^{1/2}$:

$$\frac{1}{b^2} = \frac{C}{b^{1/2}} \implies b^{3/2} = \frac{1}{C}.$$

Hence:

$$b = C^{-2/3} = \left(\frac{3h^3}{32\pi^2} \right)^{2/3} \frac{4\pi\epsilon_0}{2me^2} Z^{-1/3}.$$

Use $h = 2\pi\hbar$ and $a_0 = 4\pi\epsilon_0\hbar^2/(me^2)$:

$$b = \left(\frac{9\pi^2}{128 Z} \right)^{1/3} a_0 \approx 0.885 Z^{-1/3} a_0.$$

(d)

Continuum Fermi-gas requires local Fermi wavelength $\ll$ scale on which V varies. Fails near the nucleus ($V \rightarrow -\infty$, density diverges), in the outer fringe (few electrons, statistics breaks), and for light atoms (low N). Reliable for the core of heavy atoms.

(e)

In a screened (non-Coulombic) central field, low- ℓ orbitals *penetrate* the core and feel a larger Z_{eff} . Centrifugal barrier $\ell(\ell+1)\hbar^2/(2mr^2)$ excludes $3d$ ($\ell=2$) from the core, so $3d$ sees $Z_{\text{eff}} \approx 1$. The $4s$ orbital, although higher n , has substantial inner radial probability and sits at lower energy than $3d$. Therefore K is $[\text{Ar}]4s^1$.

Question 2 — Hyperfine structure in magnetic fields

Problem. Hamiltonian $\mathcal{H} = A\vec{I} \cdot \vec{J} + g_J\mu_B\vec{J} \cdot \vec{B} - g_I\mu_N\vec{I} \cdot \vec{B}$. (a) interpret each term; (b) estimate weak-field threshold; (c) derive $\Delta E = \frac{1}{2}AK + g_F\mu_B B M_F$; (d) draw level diagram for $J = 3/2, I = 3$; (e) strong-field hydrogen $1s^2S_{1/2}$, verify ratio.

(a)

$A\vec{I} \cdot \vec{J}$: magnetic hyperfine coupling between nuclear moment $\propto \vec{I}$ and the electronic field at the nucleus.

$g_J\mu_B\vec{J} \cdot \vec{B}$: electronic Zeeman coupling of $-g_J\mu_B\vec{J}/\hbar$ to $\vec{B}$.

$-g_I\mu_N\vec{I} \cdot \vec{B}$: nuclear Zeeman coupling; smaller than the electronic term by $\mu_N/\mu_B \sim m_e/m_p \approx 1/1836$.

(b)

Weak field requires hyperfine $\gg$ electronic Zeeman:

$$B \ll B^* \equiv \frac{AI}{g_J\mu_B}.$$

Use alkali ground-state $A/\hbar \sim 1$ GHz, $g_J \approx 2$, $I \sim 3/2$, $\mu_B = 9.27 \times 10^{-24}$ J/T:

$$A \approx \hbar \cdot 10^9 \text{ Hz} \approx 6.6 \times 10^{-25} \text{ J}.$$

Substitute:

$$B^* \sim \frac{6.6 \times 10^{-25} \cdot 1.5}{2 \cdot 9.27 \times 10^{-24}} \text{ T} \approx 0.05 \text{ T}.$$

$B \lesssim 50 \text{ mT for weak-field regime.}$

(c)

Form $\vec{F} = \vec{I} + \vec{J}$, $F \in \{|I - J|, \dots, I + J\}$. From $\vec{F}^2 = \vec{I}^2 + \vec{J}^2 + 2\vec{I} \cdot \vec{J}$:

$$\vec{I} \cdot \vec{J} = \frac{1}{2}[F(F+1) - I(I+1) - J(J+1)]\hbar^2.$$

Absorb $\hbar^2$ into A :

$$\langle A\vec{I} \cdot \vec{J} \rangle = \frac{1}{2}AK, \quad \boxed{K = F(F+1) - I(I+1) - J(J+1).}$$

Treat Zeeman as perturbation in $|F, M_F\rangle$. Project (Landé):

$$\langle \vec{J} \rangle = \frac{\vec{F} \cdot \vec{J}}{F(F+1)\hbar^2} \vec{F}, \quad \langle \vec{I} \rangle = \frac{\vec{F} \cdot \vec{I}}{F(F+1)\hbar^2} \vec{F}.$$

Compute the dot products:

$$\vec{F} \cdot \vec{J} = \frac{1}{2}[F(F+1) + J(J+1) - I(I+1)]\hbar^2,$$

$$\vec{F} \cdot \vec{I} = \frac{1}{2}[F(F+1) + I(I+1) - J(J+1)]\hbar^2.$$

Take $\vec{B} = B\hat{z}$, $\langle F_z \rangle = M_F\hbar$:

$$\Delta E_Z = g_J\mu_B B \langle J_z \rangle - g_I\mu_N B \langle I_z \rangle.$$

Drop the nuclear piece ($\mu_N/\mu_B \ll 1$):

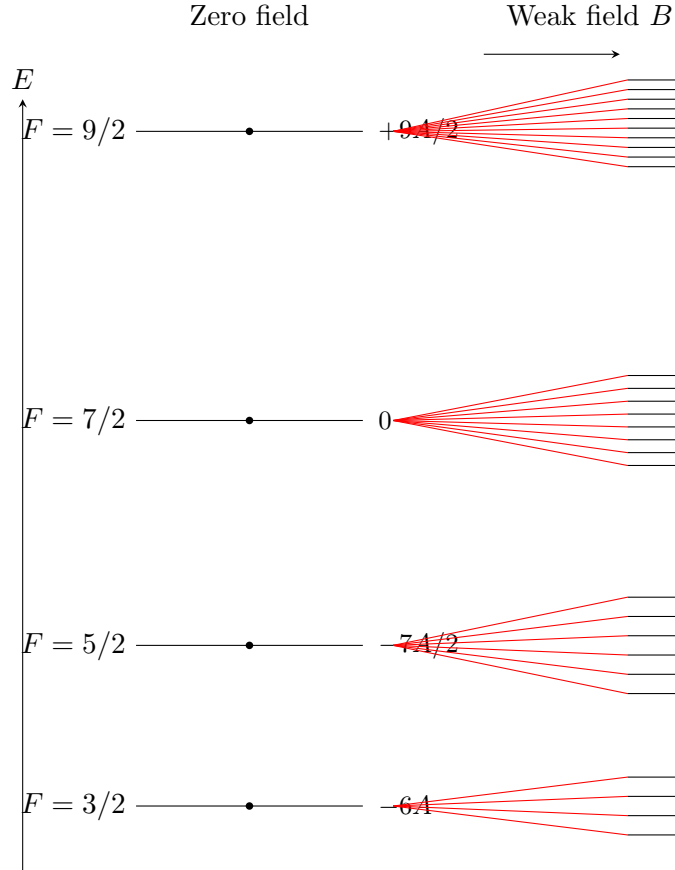
$$\boxed{g_F = g_J \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}, \quad \Delta E = \frac{1}{2}AK + g_F\mu_B B M_F.}$$

(d)

For $J = 3/2, I = 3$: $F = 3/2, 5/2, 7/2, 9/2$. Compute $K = F(F+1) - 12 - 15/4$:

F	$F(F+1)$	K	$\frac{1}{2}AK$
3/2	15/4	-12	-6A
5/2	35/4	-7	-7A/2
7/2	63/4	0	0
9/2	99/4	+9	+9A/2

Each F splits into $2F+1$ Zeeman sublevels: $4+6+8+10=28=(2J+1)(2I+1)$. Sublevel spacing $g_F\mu_B B$ depends on F .



(e)

Strong field: J_z, I_z good. For $1s^2S_{1/2}$ of H, $J = S = 1/2$, $I = 1/2$, $g_J \rightarrow g_s$, $g_I \rightarrow g_p$. First-order shifts:

$$\Delta E(M_J, M_I) = g_s \mu_B B M_J - g_p \mu_N B M_I + A M_J M_I.$$

Tabulate ($M_J, M_I = \pm 1/2$):

(M_J, M_I)	ΔE
$(+\frac{1}{2}, +\frac{1}{2})$	$\frac{1}{2}g_s\mu_B B - \frac{1}{2}g_p\mu_N B + \frac{1}{4}A$
$(+\frac{1}{2}, -\frac{1}{2})$	$\frac{1}{2}g_s\mu_B B + \frac{1}{2}g_p\mu_N B - \frac{1}{4}A$
$(-\frac{1}{2}, +\frac{1}{2})$	$-\frac{1}{2}g_s\mu_B B - \frac{1}{2}g_p\mu_N B - \frac{1}{4}A$
$(-\frac{1}{2}, -\frac{1}{2})$	$-\frac{1}{2}g_s\mu_B B + \frac{1}{2}g_p\mu_N B + \frac{1}{4}A$

Form numerator (electronic Zeeman cancels for opposite M_J):

$$\Delta E_{+,-} + \Delta E_{-,-} = g_p \mu_N B.$$

Form denominator (nuclear Zeeman cancels for opposite M_I):

$$\Delta E_{+,-} + \Delta E_{+,+} = g_s \mu_B B.$$

Divide:

$$\frac{\Delta E_{J_z=1/2, I_z=-1/2} + \Delta E_{J_z=-1/2, I_z=-1/2}}{\Delta E_{J_z=1/2, I_z=-1/2} + \Delta E_{J_z=1/2, I_z=1/2}} = \frac{g_p \mu_N}{g_s \mu_B}.$$

Question 3 — Saturated gain and a spherical astrophysical maser

Problem. Two-level homogeneously broadened gain medium with densities $N_{1,2}$, degeneracies $g_{1,2}$, pump rates $R_{1,2}$, lifetimes $\tau_{1,2}$, lineshape $g(\omega - \omega_0)$, monochromatic laser $I(z, \omega) = I(z)\delta(\omega - \omega_L)$. (a) derive $dI/dz = \alpha_0 I / (1 + I/I_S)$ with α_0 and I_S in terms of Einstein coefficients; (b) for a uniformly inverted sphere, derive the radial growth equation; (c) given $I(R) = 0.1I_S$, show $P(r) = 4\pi r^2 I(r)$ grows exponentially and find $I(r)$; (d) show $I(r)$ has a minimum, find its radius; (e) comment on $I(r = 0)$.

(a)

Net stimulated rate (gain – absorption) per unit volume per unit frequency:

$$W_{\text{stim}} = [N_2 B_{21} - N_1 B_{12}] u(\omega) g(\omega - \omega_0), \quad B_{12} g_1 = B_{21} g_2.$$

For monochromatic $I(z, \omega) = I(z)\delta(\omega - \omega_L)$ with $u = I/c$:

$$\frac{dI(z)}{dz} = \hbar\omega_0 \left[N_2 - \frac{g_2}{g_1} N_1 \right] B_{21} g(\omega_L - \omega_0) \frac{I(z)}{c}.$$

Define small-signal gain at line centre with unsaturated populations $N_{i,0} = R_i \tau_i$:

$$\alpha_0 = \frac{\hbar\omega_0}{c} \left[N_{2,0} - \frac{g_2}{g_1} N_{1,0} \right] B_{21} g(\omega_L - \omega_0).$$

Rate equations under stimulated drive (intensity I , monochromatic):

$$\begin{aligned} \dot{N}_2 &= R_2 - \frac{N_2}{\tau_2} - \left[N_2 - \frac{g_2}{g_1} N_1 \right] \frac{B_{21} I}{c} g(\omega_L - \omega_0), \\ \dot{N}_1 &= R_1 - \frac{N_1}{\tau_1} + \left[N_2 - \frac{g_2}{g_1} N_1 \right] \frac{B_{21} I}{c} g(\omega_L - \omega_0). \end{aligned}$$

Set steady state $\dot{N}_{1,2} = 0$ and let $\Delta N = N_2 - (g_2/g_1)N_1$. Solve linearly:

$$\Delta N = \frac{\Delta N_0}{1 + I/I_S}, \quad \Delta N_0 = R_2 \tau_2 - \frac{g_2}{g_1} R_1 \tau_1.$$

Read off the saturation intensity (collecting the prefactor multiplying I):

$$\alpha_0 = \frac{\hbar\omega_0}{c} \Delta N_0 B_{21} g(\omega_L - \omega_0), \quad I_S = \frac{c}{B_{21} g(\omega_L - \omega_0) [\tau_2 + (g_2/g_1)\tau_1]}.$$

Insert back:

$$\frac{dI(z)}{dz} = \frac{\alpha_0 I(z)}{1 + I(z)/I_S}.$$

(b)

Power crossing radius r : $P(r) = 4\pi r^2 I(r)$. Energy balance over shell $[r, r + dr]$ with local gain $dI/d\ell = \alpha_0 I / (1 + I/I_S)$ along a ray:

$$dP = \frac{\alpha_0 I(r)}{1 + I(r)/I_S} 4\pi r^2 dr.$$

Differentiate $P = 4\pi r^2 I$:

$$\frac{dP}{dr} = 4\pi r^2 \frac{dI}{dr} + 8\pi r I.$$

Equate the two expressions and divide by $4\pi r^2$:

$$\boxed{\frac{dI}{dr} + \frac{2I}{r} = \frac{\alpha_0 I}{1 + I/I_S}.$$

(c)

Inside the sphere $I(r) \leq I(R) = 0.1 I_S \ll I_S$, so $1 + I/I_S \approx 1$ (unsaturated):

$$\frac{dI}{dr} + \frac{2I}{r} = \alpha_0 I.$$

Multiply by $4\pi r^2$ to convert to P :

$$\frac{dP}{dr} = \alpha_0 P.$$

Integrate:

$$P(r) = P(R) e^{-\alpha_0(R-r)} \iff P(r) \propto e^{\alpha_0 r}.$$

Read off $I(r) = P(r)/(4\pi r^2)$:

$$\boxed{I(r) = I(R) \left(\frac{R}{r}\right)^2 e^{-\alpha_0(R-r)} = 0.1 I_S \left(\frac{R}{r}\right)^2 e^{-\alpha_0(R-r)}.$$

(d)

Minimise: take $dI/dr = 0$. From part (c):

$$\frac{dI}{dr} = I \left(-\frac{2}{r} + \alpha_0 \right).$$

Set bracket to zero:

$$\boxed{r_{\min} = \frac{2}{\alpha_0}.$$

Second derivative is positive (factor $2/r^2 > 0$ dominates near zero), so this is a minimum.

(e)

At $r = 0$ the formula $I(r) = 0.1 I_S (R/r)^2 e^{-\alpha_0(R-r)}$ diverges as r^{-2} . Physically this is unphysical: (i) the geometric factor reflects an idealised point convergence of outward-going rays whose backward extrapolation crowds at the origin, (ii) once $I \rightarrow I_S$ the unsaturated approximation fails and saturation clamps the intensity, (iii) the spherical-wave $1/r^2$ assumption breaks below a wavelength. Take-away: in real masers the centre is heavily saturated and the actual $I(0)$ is bounded by $\sim I_S$, not infinite.

Question 4 — Semi-classical atom–field interaction

Problem. Two-level atom in classical field $V(t) = ex\mathcal{E}_0 \cos \omega t$, $\omega = \omega_0 + \delta\omega$, $|\delta\omega/\omega_0| \ll 1$. (a) derive the coupled amplitude equations; (b) RWA give $|c_2(t)|^2$; (c) derive $B_{12} = \pi\chi_{12}^2/(3\varepsilon_0\hbar^2)$; (d) compute B_{12} for $1s \rightarrow 2p$.

(a)

Expand state in unperturbed eigenstates:

$$\Psi(\vec{r}, t) = c_1(t)\psi_1 e^{-iE_1 t/\hbar} + c_2(t)\psi_2 e^{-iE_2 t/\hbar}.$$

Insert into $i\hbar\partial_t\Psi = (H_0 + V)\Psi$ and use $H_0\psi_n = E_n\psi_n$:

$$i\hbar[\dot{c}_1\psi_1 e^{-iE_1 t/\hbar} + \dot{c}_2\psi_2 e^{-iE_2 t/\hbar}] = V(t)[c_1\psi_1 e^{-iE_1 t/\hbar} + c_2\psi_2 e^{-iE_2 t/\hbar}].$$

Project on ψ_1^* and ψ_2^* (orthonormality):

$$i\hbar\dot{c}_1 = c_2 V_{12}(t) e^{-i\omega_0 t}, \quad i\hbar\dot{c}_2 = c_1 V_{21}(t) e^{+i\omega_0 t},$$

with $\omega_0 = (E_2 - E_1)/\hbar$, $V_{ab} = \int \psi_a^* V \psi_b d\vec{r}$. Substitute $V = e x \mathcal{E}_0 \cos \omega t$ and use $\chi_{ab} = -e \int \psi_a^* x \psi_b d\vec{r}$:

$$V_{ab}(t) = -\mathcal{E}_0 \cos(\omega t) \chi_{ab}.$$

Expand cosine: $\cos \omega t = \frac{1}{2}(e^{i\omega t} + e^{-i\omega t})$:

$$\begin{cases} \dot{c}_1(t) = \frac{i\mathcal{E}_0\chi_{12}}{2\hbar} [e^{i(\omega-\omega_0)t} + e^{-i(\omega+\omega_0)t}] c_2(t), \\ \dot{c}_2(t) = \frac{i\mathcal{E}_0\chi_{12}}{2\hbar} [e^{-i(\omega-\omega_0)t} + e^{i(\omega+\omega_0)t}] c_1(t). \end{cases} \quad (1)$$

Used $\chi_{21} = \chi_{12}^* = \chi_{12}$ (real ψ_n).

(b)

Initial $c_1(0) = 1$, $c_2(0) = 0$. Weak field: set $c_1 \approx 1$. Drop the counter-rotating $e^{i(\omega+\omega_0)t}$ term (RWA, since $\omega + \omega_0 \sim 2\omega \gg |\omega - \omega_0|$):

$$\dot{c}_2 = \frac{i\mathcal{E}_0\chi_{12}}{2\hbar} e^{-i(\omega-\omega_0)t}.$$

Integrate from 0 to t :

$$c_2(t) = \frac{i\mathcal{E}_0\chi_{12}}{2\hbar} \frac{e^{-i(\omega-\omega_0)t} - 1}{-i(\omega - \omega_0)}.$$

Square modulus, use $|e^{i\theta} - 1|^2 = 4 \sin^2(\theta/2)$:

$$|c_2(t)|^2 = \frac{\mathcal{E}_0^2 \chi_{12}^2}{\hbar^2} \frac{\sin^2[(\omega - \omega_0)t/2]}{(\omega - \omega_0)^2}. \quad (2)$$

(c)

Replace $\mathcal{E}_0^2$ by spectral content: $\varepsilon_0 \mathcal{E}_0^2/2 = u(\omega)d\omega$ gives $\mathcal{E}_0^2(\omega) = 2u(\omega)/\varepsilon_0$. Sum incoherently:

$$|c_2(t)|^2 = \int \frac{2\chi_{12}^2 u(\omega)}{\varepsilon_0 \hbar^2} \frac{\sin^2[(\omega - \omega_0)t/2]}{(\omega - \omega_0)^2} d\omega.$$

For large t kernel sharpens:

$$\frac{\sin^2[(\omega - \omega_0)t/2]}{(\omega - \omega_0)^2} \rightarrow \frac{\pi t}{2} \delta(\omega - \omega_0).$$

Pull out slowly varying $u(\omega_0)$:

$$|c_2(t)|^2 = \frac{\pi\chi_{12}^2}{\varepsilon_0\hbar^2} u(\omega_0) t.$$

Identify rate $\equiv B_{12}u(\omega_0)$ and average $\overline{\cos^2\theta} = 1/3$ over orientation of $\vec{E}$ relative to dipole:

$$\boxed{B_{12} = \frac{\pi\chi_{12}^2}{3\varepsilon_0\hbar^2}.} \quad (3)$$

(d)

Need $\sum_m |\chi_{1s \rightarrow 2p,m}|^2$. Write $x = r \sin\theta \cos\phi$ via spherical harmonics:

$$Y_{1,-1} - Y_{1,+1} = 2\sqrt{\frac{3}{8\pi}} \sin\theta \cos\phi.$$

Solve for x :

$$x = r\sqrt{\frac{2\pi}{3}} (Y_{1,-1} - Y_{1,+1}).$$

Factor matrix element: $\chi_{1s,2p_m} = -e I_r A_m$ with

$$I_r = \int_0^\infty R_{10}(r) R_{21}(r) r^3 dr, \quad A_m = \sqrt{\frac{2\pi}{3}} \int Y_{0,0}^* (Y_{1,-1} - Y_{1,+1}) Y_{1,m} d\Omega.$$

Use $Y_{0,0} = 1/\sqrt{4\pi}$ and orthonormality with the conventions in the table. The table gives $Y_{1,-1} = (3/8\pi)^{1/2} \sin\theta e^{-i\phi}$ and $Y_{1,+1} = -(3/8\pi)^{1/2} \sin\theta e^{i\phi}$, so $\int Y_{1,-1}^* Y_{1,m} d\Omega = \delta_{m,-1}$ and $\int Y_{1,+1}^* Y_{1,m} d\Omega = \delta_{m,+1}$. Hence:

$$A_0 = 0, \quad A_{+1} = -\frac{1}{\sqrt{6}}, \quad A_{-1} = +\frac{1}{\sqrt{6}}.$$

Sum the squares:

$$\sum_m |A_m|^2 = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}.$$

Insert the radial functions $R_{10} = 2a_0^{-3/2} e^{-r/a_0}$, $R_{21} = (1/\sqrt{3}) a_0^{-5/2} r e^{-r/2a_0}$:

$$I_r = \frac{2}{\sqrt{3}} a_0^{-4} \int_0^\infty r^4 e^{-3r/(2a_0)} dr.$$

Use $\int_0^\infty r^n e^{-\alpha r} dr = n!/\alpha^{n+1}$:

$$I_r = \frac{2}{\sqrt{3}} a_0^{-4} \cdot \frac{4!}{(3/2a_0)^5}.$$

Reduce:

$$I_r = \frac{2}{\sqrt{3}} \cdot 24 \cdot \frac{2^5}{3^5} a_0 = \frac{2^8 \sqrt{3}}{3^5} a_0 = \frac{256\sqrt{3}}{243} a_0.$$

Combine into $\sum_m \chi_{12}^2$:

$$\sum_m |\chi_{1s,2p_m}|^2 = e^2 I_r^2 \sum_m |A_m|^2.$$

Substitute numerics:

$$\sum_m |\chi_{1s,2p_m}|^2 = e^2 \left(\frac{256\sqrt{3}}{243} \right)^2 a_0^2 \cdot \frac{1}{3} = e^2 a_0^2 \left(\frac{256}{243} \right)^2.$$

Insert into (3):

$$B_{12} = \frac{\pi}{3\varepsilon_0\hbar^2} \left(\frac{256}{243} \right)^2 e^2 a_0^2.$$

Plug in $e = 1.602 \times 10^{-19} \text{ C}$, $a_0 = 5.292 \times 10^{-11} \text{ m}$, $\varepsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$, $\hbar = 1.055 \times 10^{-34} \text{ J s}$.

Evaluate the dimensional ratio:

$$\frac{e^2 a_0^2}{\varepsilon_0 \hbar^2} = \frac{(1.602 \times 10^{-19})^2 (5.292 \times 10^{-11})^2}{(8.854 \times 10^{-12})(1.055 \times 10^{-34})^2}.$$

Reduce step by step:

$$e^2 = 2.566 \times 10^{-38} \text{ C}^2.$$

$$a_0^2 = 2.800 \times 10^{-21} \text{ m}^2.$$

$$e^2 a_0^2 = 7.185 \times 10^{-59} \text{ C}^2 \text{ m}^2.$$

$$\varepsilon_0 \hbar^2 = (8.854 \times 10^{-12})(1.113 \times 10^{-68}) = 9.857 \times 10^{-80}.$$

$$\frac{e^2 a_0^2}{\varepsilon_0 \hbar^2} = \frac{7.185 \times 10^{-59}}{9.857 \times 10^{-80}} = 7.29 \times 10^{20} \text{ m}^3 \text{ J}^{-1} \text{ s}^{-2}.$$

Compute $(256/243)^2$:

$$\left(\frac{256}{243} \right)^2 = 1.110.$$

Combine:

$$B_{12} = \frac{\pi}{3} \cdot 1.110 \cdot 7.29 \times 10^{20}.$$

Multiply:

$$B_{12}^{1s \rightarrow 2p} \approx 8.5 \times 10^{20} \text{ m}^3 \text{ J}^{-1} \text{ s}^{-2}.$$